A finite-seed parameter refinement for one-sample matroid prophet inequalities

Preparatory research note, October 10, 2026

Abstract

We give a candidate refinement of the constants in OpenAI's one-sample matroid prophet construction. Retaining its combinatorial machinery, we derive a coefficient greater than 2^{-63} against an adversary that sees every sample, online value, and seed bit. The accessible source already derives 2^{-293}, stronger than its advertised 2^{-310}. The changes are a sharper use of existing exit and residual estimates, affine mixing of branches, and sixteen shared rounding offsets. An explicit implementation uses $62n + 40$ fair bits; no polynomial running time is asserted. Exact scalar checks and finite online-rule checks accompany the derivation. This is an AI-assisted, unreviewed draft, not a formalized proof. Daniel's October 8 announcement of 2^{-47} is stronger; its proof was unavailable in our audit. We make no claim to the strongest current result or to priority.

1 Model, claim, and provenance

Let $M = (E, \mathcal{I})$ be a known finite labeled matroid, $n = |E|$, with a fixed total label order. For $A, P \subseteq E$, write $\mathrm{r}(A \mid P) = \mathrm{r}(A \cup P) - \mathrm{r}(P)$ and $L = \mathrm{cl}(\varnothing)$. For a nonnegative vector w, put $\mathrm{OPT}_M(w) = \max_{I \in \mathcal{I}} \sum_{e \in I} w_e$.

Proposed theorem 1. *For every such M there is a measurable one-sample online rule whose seed consists of $62n + 40$ independent fair bits and whose law does not depend on the value distributions. Let all $2n$ variables S_e, V_e be mutually independent, nonnegative, with S_e and V_e identically distributed for each e. Let the seed R be independent of them and assume $\mathbb{E}\,\mathrm{OPT}_M(V) < \infty$. For every measurable arrival permutation $\pi(S, V, R)$, the rule accepts an independent set at every prefix and its reward satisfies*

$$\mathbb{E}\,\mathrm{reward} \;\geq\; 2^{-63}\mathbb{E}\,\mathrm{OPT}_M(V).$$

The rule receives all samples initially, uses no distribution description, and observes online values only upon their arrival. The adversary may observe the entire initial seed, including unused coins.

The argument below depends on named combinatorial lemmas of the September 23 manuscript [1], pinned at commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. We restate the interfaces needed for retuning and derive all changed quantitative estimates. We do not claim a new density decomposition, marked-pivot theorem, reverse-exposure method, or sample/value reduction. Source arguments, rather than their hard-coded numerical corollaries, are the dependencies. The source's final calculation is $2^{-140-16-5-100-32} = 2^{-293}$ and its sample/value reduction is lossless.

The stronger 2^{-47} public announcement [2] must be resolved before any best-known-bound claim. A separate $1/2$ result [3] uses an order fixed independently of samples and values; it does not supply the full-seed quantifier above. The draft and the source dependency audit received a

second pass by the same agent, not independent mathematical review. No Lean build certifies this draft.

2 A fully specified fixed-vector rule

First fix $w \in [0,\infty)^E$ independently of the seed. A rule reveals and sacrifices a seed-selected mask, then receives all labels in an arbitrary order. Its target is $\mathbb{E}_R \min_\pi w(A(w,R,\pi))$. Choose

$$B = 2^{16}, \quad \kappa = 2^{18}, \quad t = 2^{-29}, \quad p = 1 - 2^{-35}, \quad \beta = 1 - B^{-1}. \tag{1}$$

Choose J uniformly in $\{0,\dots,15\}$, and round positive coordinates by

$$u_J(x) = 2^J B^{\lfloor \log_B(x/2^J) \rfloor}, \qquad u_J(0) = 0.$$

Use the same J for every coordinate. The branch is main with probability p and maximum otherwise. All masks, parity, branch, and offset coins are independent and drawn initially.

The maximum branch reveals a fair mask and accepts the first unmasked positive nonloop with priority exceeding every revealed positive nonloop, then stops. Higher rounded weight means higher priority, with the fixed label order resolving ties. An empty revealed threshold is below every positive nonloop. If m_* is the greatest rounded nonloop weight, this branch has expected worst-order reward at least $m_*/4$: it succeeds whenever the highest-priority nonloop is unmasked and the second is masked. With just one positive nonloop it succeeds with probability $1/2$.

In the main branch draw masks H, D, C, T with respective rates $1/2, 1/4, t, t$, independently across labels. Reveal and sacrifice $H \cup D \cup C$. Define the analytical candidate set

$$U = \{e \notin H : u_e > 0,\ e \notin \mathrm{cl}(H_{>e})\}, \qquad Y = I^* \setminus H,$$

where I^* is priority greedy on the whole rounded vector and $H_{>e}$ contains the higher-priority labels in H. The candidate test uses only the revealed H values and the value at the tested label. The source's candidate-mass lemma gives

$$Y \subseteq U, \qquad \mathbb{E}_H u(Y) = W/2, \qquad \mathbb{E}_H u(U) \le W, \qquad W = \mathrm{OPT}_M(u). \tag{2}$$

Indeed the probability contribution of a candidate equals that of a label chosen by greedy on H, conditional on its predecessors in H.

For a fixed offset divide weights by 2^J in the analysis, so levels are B^i. Put $U_i = \{e \in U : u_e = B^i\}$, $n_i = |U_i|$, $Y_i = Y \cap U_i$. List only levels intersecting D, in increasing weight order, as $G_1,\dots,G_m$. Write $D_h = D \cap G_h$ and likewise for C_h, T_h. Their revealed parts D_h, C_h suffice for preparation; the full groups are analysis objects.

For a flat P, let $Q_h(P)$ be the largest maximizing flat of $|D_h \cap Z| - \kappa\, \mathrm{r}(Z)$ over flats $Z \supseteq P$. For nonflat arguments first take closure. Set $a_h = -3h$ and let $Q_{h,k} = Q_h$ for $k \ge a_h$, the identity otherwise. Starting from $F_0(k) = L$ for $k < 0$ and $F_0(k) = E$ otherwise, set

$$X_h(k) = Q_{h,k}(F_{h-1}(k)), \qquad F_h(k) = \mathrm{cl}\big(X_h(k) \cup (C_h \cap X_h(k+1))\big). \tag{3}$$

These paths are computed in increasing h. All relevant times lie in a finite interval of length $O(n)$. Let $\widehat{F} = F_m$; reject everything if $m = 0$. Draw a fair parity $\xi \in \{0,1\}$.

For an arriving label e, reject sacrificed labels, failed candidates, and unlisted levels. For its listed index h, find its nominal birth $b_h(e) = \min\{k : e \in X_h(k)\}$. It is eligible precisely when

$$e \in T, \quad b_h(e) \ge a_h + 2, \quad b_h(e) \equiv \xi \pmod 2, \quad e \notin \widehat{F}(b_h(e) - 2).$$

At each endpoint b maintain a set A_b. Accept an eligible e of birth b exactly when $\mathrm{r}(\{e\} \mid \widehat{F}(b-2) \cup A_b) = 1$.

The source density and path lemmas hold for every $\kappa > 0$: Q_h is increasing and extensive, and

$$|D_h \cap (\mathrm{cl}(Q_h(P) \cup Z) \setminus Q_h(P))| \le \kappa\, \mathrm{r}(Z \mid Q_h(P)), \tag{4}$$

$$F_{h-1}(k) \subseteq X_h(k) \subseteq F_h(k) \subseteq X_h(k+1). \tag{5}$$

They follow from supermodularity of the density objective on flats and from the path recursion. Nonloops are born at a_h or in $[a_h+2, 0]$. Each path has generators $J_h \subseteq D_h$ with increasing subsets $J_h(k)$ such that

$$X_h(k) = \mathrm{cl}(F_{h-1}(k) \cup J_h(k)), \qquad |J_h| \le |D_h|/\kappa. \tag{6}$$

The rank costs telescope as new density generators are inserted. These facts require no large lower bound on κ. By (5), successive layer bases contain all earlier layers, proving independence at every arrival prefix. Finite rank operations, fixed tie rules, and countably many measurable rounding levels prove measurability.

3 Safety with a larger exit cutoff

Fix H, D and a listed index h. Define

$$K_{h,b} = \bigcup_{j<h} \big(T_j \cap (X_j(b) \setminus X_j(b-1))\big).$$

It contains all lower-weight eligible competitors in layer b and is contained in $F_{h-1}(b)$. A label $d \in D_h \cap Y$ is safe if its birth b is in $[a_h + 2, 0]$, has parity ξ, and $d \notin Q_h(F_{h-1}(b-2) \cup K_{h,b})$. Let σ_h count them.

The source's reverse-exposure construction queries departing guard coordinates once each, in predictable batches. Conditional on a transcript just before a step, the surviving coordinates retain their product law. If a nonloop is still present, let q_{b-1} be its conditional exit probability at the next reverse step; put it zero after exit. The disjoint-query-certificate lemma bounds the expected number of survivals in a hazard band $[q_0, 2q_0)$ by $\log(1/q_0)$. Adding the possible exit step and summing dyadic bands therefore gives, for an integer cutoff exponent m_0,

$$\mathbb{P}(q_{\mathrm{exit}} < 2^{-m_0}) \le 2^{1-m_0}((m_0+2)\log 2 + 1). \tag{7}$$

This is the source argument with its final cutoff left symbolic. It uses a finite query horizon and an increasing Boolean event; it does not use the original numerical t, B, κ.

The positive series $\log 2 = 2\sum_{j\ge 0}((2j+1)3^{2j+1})^{-1}$ implies $2/3 < \log 2 \le 25/36 < 7/10$. For the upper bound, keep the first term and bound subsequent denominators $2j+1$ by 3. Taking $m_0 = 4$ in (7) gives a failure probability at most $13/20$. Every nonloop exits, so

$$\sum_b \mathbb{E}[\mathbf{1}_{\{d \in X_h(b)\}} q_{b-1}^2 \mid H, D] = \mathbb{E}[q_{\mathrm{exit}} \mid H, D] \ge \frac{7}{320}. \tag{8}$$

The source hybrid transition replaces just the departing guard bits by their independent test bits. Conditional on the complete ordinary transcript, unqueried guard bits and these test bits jointly have the original product law. Since guard and test rates agree, the hybrid exit has conditional probability q_{b-1}, and the probability of ordinary and hybrid exit together is $\mathbf{1}_{\{d \in X_h(b)\}} q_{b-1}^2$ given the pre-step transcript. The hybrid output contains $Q_h(F_{h-1}(b-2) \cup K_{h,b})$ for valid b; hence

double exit implies safety before parity. This is an endpointwise argument, not an assertion that different endpoint experiments are independent.

The only excluded birth is a_h; none occurs at a_h+1. Since Y is independent, at most $\mathrm{r}(Q_h(L)) \le |D_h|/\kappa$ of its sampled labels have this birth. Subtract this loss from (8) and then average fair parity:

$$\mathbb{E}[\sigma_h \mid H, D] \ge \frac{7}{640}|D_h \cap Y| - \frac{|D_h|}{2\kappa}.$$

Extend σ_i by zero on unlisted true groups. Averaging D without conditioning on listing gives

$$\mathbb{E}[\sigma_i \mid H] \ge \gamma|Y_i| - \frac{n_i}{8\kappa}, \qquad \gamma = \frac{7}{2560}. \tag{9}$$

4 Residual entropy and nominal rank

Let $\mathcal{B}_h = \{b : a_h + 2 \le b \le 0,\ b \equiv \xi \pmod 2\}$. For $b \in \mathcal{B}_h$ put

$$P_{h,b} = G_h \cap (X_h(b) \setminus X_h(b-1)), \quad O = D \cup C, \quad S_{h,b} = P_{h,b} \setminus O,$$

and set $P_{h,b} = \varnothing$ for other endpoints of this parity. The nominal rank is

$$z_h = \sum_b \mathrm{r}(S_{h,b} \mid X_h(b-2) \cup K_{h,b}).$$

Choose a label-ordered basis extension $Z_b \subseteq S_{h,b}$ for each summand, so $\sum_b |Z_b| = z_h$. The residual

$$\mathcal{R}_h = \bigcup_b (P_{h,b} \setminus \mathrm{cl}(X_h(b-2) \cup K_{h,b} \cup Z_b))$$

is contained in O. Applying (4) with the larger base $Q_h(F_{h-1}(b-2) \cup K_{h,b})$ gives the deterministic inequality

$$\sigma_h \le \kappa z_h + |\mathcal{R}_h|. \tag{10}$$

We use the following precise interface of the source marked-pivot and residual-family lemmas. Conditional on H, parity, and all D, C, T bits outside a true group U_i of size $n \ge \kappa$, there is a fixed family $\mathfrak{R}_i$ containing every residual on every listed outcome with $z_h \le n/\kappa$. For $q = \lfloor n/\kappa \rfloor$ and $S_d(n) = \sum_{\ell=0}^{d} \binom{n}{\ell}$, it satisfies

$$|\mathfrak{R}_i| \le S_q(n)^2 \big(4^{2q} S_{2q}(n)\big)^2. \tag{11}$$

No focal bit or listing event is in this conditioning. The family allows at most q density generators and q witness labels; two marked-pivot sequences encode valid births and failure of spanning. The source marked-pivot lemma bounds s movable occurrences by $4^s S_s(n)$ regardless of the timeline length. That combinatorial lemma, including its matroid and hyperplane-arrangement proof, remains an external dependency of this note.

For $0 < \alpha \le 1/2$, $\alpha^{\alpha n} S_{\lfloor \alpha n \rfloor}(n) \le (1+\alpha)^n \le e^{\alpha n}$. Use $\alpha = 1/\kappa, 2/\kappa$ in (11), keeping the floors, to obtain

$$\log |\mathfrak{R}_i| \le \frac{n}{\kappa}(6 \log \kappa + 6 + 4 \log 2).$$

This sharper expression is already inside the source proof, before its relaxation to $1000 \log \kappa$. For $\kappa = 2^{18}$ and $\log 2 > 2/3$, it implies $|\mathfrak{R}_i| \le 2^{121 n/\kappa}$.

Under precisely the conditioning above, each focal coordinate belongs to O independently with probability $1/4 + 3t/4 \le 1/2$. Set $\delta = 139/\kappa$. A union bound gives

$$\mathbb{P}(\exists R_0 \in \mathfrak{R}_i : R_0 \subseteq O,\ |R_0| > \delta n) \le 2^{-18 n/\kappa} \le 2^{-18}.$$

Set $e_i = |\mathcal{R}_h|$ on listed outcomes with $z_i \le n/\kappa$, and zero otherwise. Then $\mathbb{E}[e_i \mid H] \le (\delta + 2^{-18})n = 140n/\kappa$. Extend z_i by zero on unlisted groups. The inequality $\sigma_i \le \kappa z_i + e_i$ holds also on unlisted outcomes and when $z_i > n/\kappa$, since $\sigma_i \le n$. Together with (9), this proves, for $n_i \ge \kappa$,

$$\mathbb{E}[z_i \mid H] \ge \frac{\gamma|Y_i| - \epsilon n_i}{\kappa}, \qquad \epsilon = \frac{1121}{2^{21}}. \tag{12}$$

5 Order-uniform accounting and branch mixing

Define

$$\lambda_h = \sum_b \mathrm{r}(T_h \cap S_{h,b} \mid \widehat{F}(b-2) \cup K_{h,b}).$$

With all coins fixed, the source order-uniform rank lemma gives $N_{\ge h}(\pi) \ge \lambda_h$ for every permutation, where $N_{\ge h}$ counts accepts of weight at least that of G_h. Indeed, the final greedy set of a layer spans its eligible labels; its selected lower weight labels lie in $K_{h,b}$. Contracting them proves the bound.

One common set of generators for the later paths is $J' = C_h \cup \bigcup_{j>h}(C_j \cup J_j)$, of size at most

$$c_h = |C_h| + \sum_{j>h}(|C_j| + |D_j|/\kappa).$$

It satisfies $X_h(k) \subseteq \widehat{F}(k) \subseteq \mathrm{cl}(X_h(k) \cup J')$ simultaneously for all k. The resulting rank losses at chosen-parity endpoints telescope, giving

$$\sum_b \mathrm{r}(S_{h,b} \mid \widehat{F}(b-2) \cup K_{h,b}) \ge z_h - c_h.$$

Specifically each loss is at most $\mathrm{r}(J' \mid X_h(b-2)) - \mathrm{r}(J' \mid X_h(b))$, so the same generators are charged only once.

Condition on H, D, C, ξ and all T bits outside G_h. The summand sets and bases are fixed and focal T remains an independent rate-t mask. Intersecting a fixed basis extension with T_h shows $\mathbb{E}[\lambda_h \mid \cdot] \ge t(z_h - c_h)$. Extend λ_i, c_i by zero on unlisted true groups. Bounding costs by counts on all true groups before averaging gives

$$\mathbb{E}[c_i \mid H] \le t n_i + (t + 1/(4\kappa)) \sum_{\ell > i} n_\ell.$$

Thus (12) implies

$$\mathbb{E}[\lambda_i \mid H] \ge \frac{t}{\kappa}\left[\gamma|Y_i| - (\epsilon + t\kappa)n_i - (t\kappa + 1/4)\sum_{\ell>i} n_\ell\right] \quad (n_i \ge \kappa). \tag{13}$$

No chosen arrival order has been conditioned upon.

Each accepted weight B^ℓ is counted in cumulative level counts by at most $\sum_{i \le \ell} B^i = B^\ell/\beta$. Consequently

$$\min_\pi u(A_{\mathrm{main}}(\pi)) \ge \beta \sum_{i : n_i \ge \kappa} B^i \lambda_i. \tag{14}$$

Also

$$\sum_i B^i \sum_{\ell > i} n_\ell \le \frac{u(U)}{B-1}, \qquad \sum_{i : n_i < \kappa} B^i |Y_i| \le \frac{\kappa m_*}{\beta}.$$

Multiply (13) by levels, use (14), and average (2). Restoring the factor 2^J when needed gives

$$\mathbb{E}\min_{\pi} u(A_{\text{main}}(\pi)) \geq AW - t\gamma m_*, \qquad A = \frac{\beta t}{\kappa} s, \tag{15}$$

where

$$s = \frac{\gamma}{2} - \epsilon - t\kappa - \frac{t\kappa + 1/4}{B-1} = \frac{3120271}{9162457088} > \frac{1}{3072}. \tag{16}$$

The affine bound may be negative for a particular vector; that causes no problem. Mix in the maximum branch:

$$\mathbb{E}\min_{\pi} u(A(\pi)) \geq pAW + \left(\frac{1-p}{4} - pt\gamma\right) m_* \geq pAW,$$

because $(1-p)/4 = t/256$ and $\gamma = 7/2560 < 1/256$. Both minima occur before averaging the respective seeds, so the adversary may select its order after seeing the branch.

6 Finite offsets and lossless one-sample transfer

For any $x > 0$, let $k = \lfloor \log_2 x \rfloor$. As J ranges over the sixteen phases, rounded exponents are exactly $k, k-1, \ldots, k-15$. Therefore

$$\mathbb{E}_J u_J(x) = \frac{2^{k+1}\beta}{16} \geq \frac{\beta}{16} x.$$

The same inequality holds at zero. Fix an optimum independent set for w before drawing J, and sum to get $\mathbb{E}_J \operatorname{OPT}_M(u_J) \geq \beta \operatorname{OPT}_M(w)/16$. Conditional on any phase the remaining seed has its original product law. True accepted weights dominate rounded weights, so

$$\mathbb{E}_{J,R}\min_{\pi} w(A(J,R,\pi)) \geq C \operatorname{OPT}_M(w), \qquad C = \frac{p\beta^2 ts}{16\kappa} > 2^{-63}. \tag{17}$$

Indeed $p\beta^2 > 3/4$, $s > 1/3072$, and $t/\kappa = 2^{-47}$, whose product in (17) is 2^{-63}. The exact coefficient is

$$C = \frac{10539177666821697366742 5}{69689828745408197317299119602026129706188 8} \approx 1.39485 \cdot 2^{-63}.$$

Visibility of the phase does not change this argument: the minimum over orders was already taken at each complete seed.

For the prophet rule, let $B_0(R)$ be its sacrificed mask and define $w_e^{\text{glue}} = S_e$ on $B_0(R)$ and V_e elsewhere. Conditional on the complete seed, this vector has the product law of V, independent of that seed. Run the hidden-vector rule using samples only on B_0, reject those labels, and use arriving values on its complement. For each S, V, R its reward under any adversarial order is at least the minimum over all permutations on the glued vector. Integrating (17) proves the proposed theorem. Unused samples and values may influence the adversary without affecting this lower bound. Feasibility bounds actual reward by $\operatorname{OPT}_M(V)$; nonnegative integration needs no extra moment hypothesis on loop values.

Predraw four phase bits, 35 branch bits (choose maximum only when all are zero), one parity bit, and per label one H, two D, 29 C, 29 T, and one maximum-mask bit. This totals $62n + 40$ fair bits. The finite rule can precompute rank and closure tables over all subsets and enumerate flats for each density map. That implementation may use exponential work, queries, and storage. Arbitrary real observations remain exact in the source measurable model; rational implementations pay input-bit costs for exact level comparisons. Output work is at least the number of accepted labels. No efficiency or precision improvement is asserted. Empty ground sets, rank zero, loops, ties, zero values, and arbitrary integer weight exponents are included.

7 Evidence and release status

The accompanying standard-library Python certificate recomputes every scalar displayed above with exact rational arithmetic. Separate checks enumerate all labeled matroids on up to four elements for candidate, density, maximum-branch, and rounding facts. An online-rule checker enumerates small complete seed states, takes the worst arrival order before averaging, and checks prefix feasibility, path generators, conditional safety, later-path rank cost, and order-uniform rank bounds. Small density thresholds and a larger test rate exercise nontrivial transitions; those tests do not measure performance at the proposed parameters or prove a universal matroid theorem.

The quantitative derivation has been audited against the source by the same agent that wrote it. Remaining review targets include the parameter-uniform reverse/hybrid argument and the source residual encoding under its precise conditioning. A separate analytic note shows that further searching our conservative scalar-certificate family cannot reach the announced 2^{-47}; this is a limitation of that certificate, not an impossibility theorem for the algorithm or problem. This version is preparatory work in the current repository. Authorship, independent review, novelty comparison, and a public release decision remain to be settled before a standalone announcement.

References

[1] OpenAI. *One Sample Suffices for Matroid Prophet Inequalities against an Almighty Adversary.* September 23, 2026 manuscript, retrieved at commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. Pinned manuscript. Relevant source lemma names: Candidate mass; Density expansion; Path properties; A generator budget for the entire path; Disjoint-query certificates; Exit mass; Safe sampled labels; Marked pivots; A uniform family of residuals; Order-uniform rank bound; Telescoping rank cost; Independent test thinning; One-sample simulation.

[2] Daniel (@firsttimelifer). Public announcement of a 2^{-47} guarantee, October 8, 2026. Proof and exact assumptions unavailable in this audit.

[3] Abdi, Banihashem, Hajiaghayi, and Mittal. *On the Strong Matroid Secretary Conjecture and Beyond.* arXiv:2609.19118v1, September 16, 2026. Theorem 1.3 and its fixed-order model.